Basic Info

- **Yang Rong** · Male
  - Senior Blockchain Protocol Engineer
- **Job Preferences:** Blockchain / Web3 Industry · Remote · Full-time or Part-time
- GitHub · Notes · yrong1997@gmail.com

Professional Summary

Senior protocol engineer with 5+ years of production blockchain work in Rust, Solidity, Go, and TypeScript.

Core developer since 2021 of Snowbridge, a trustless Ethereum ↔ Polkadot bridge secured by on-chain light clients on both sides. Now building speculative messaging between parachains in Polkadot-SDK.

Earlier DeFi work at Parallel Finance: a money market and AMM on Substrate (Rust), and an NFT lending protocol on Ethereum (Solidity).

Technical Skills

- **Languages:** Rust, Solidity, Go, TypeScript
- **Blockchain:** Polkadot-SDK / Substrate (FRAME, XCM) · Ethereum and L2s (Arbitrum, Base) · upgradeable contracts · Foundry, Hardhat
- **Consensus & Cryptography:** light clients (Ethereum sync committee, BEEFY) · SSZ, Merkle and MMR proofs · Fiat-Shamir sampling
- **Distributed Systems:** relayers · idempotent delivery · finality-gated processing · indexing and reconciliation
- **DeFi:** lending, liquidation, AMMs, oracles, liquid staking
- **Tooling:** PostgreSQL, GraphQL, Docker, Nix, GitHub Actions

Working Experience

- **Snowbridge (Snowfork)** | Core Developer · 2021 – Present — *Trustless bridge between Ethereum (and L2s) and Polkadot*
  - **How it works:** No multisig or external validators. Each chain verifies the other's consensus with an on-chain light client, so the bridge is as secure as the two chains themselves.
  - **Light clients:** Develop and maintain the Ethereum beacon light client (Rust). Work on the BEEFY client (Solidity), which checks a random sample of validator signatures. It has an interactive three-transaction protocol seeded by RANDAO, and a non-interactive one-transaction Fiat-Shamir mode that I shipped.
  - **Messaging:** Work on the message queues (Rust) and the Gateway V2 upgrade (Solidity), which adds arbitrary contract calls and L2 routes.
  - **Off-chain services:** Work on the relayers (Go), including the fisherman that reports equivocating validators, and maintain the indexer and monitoring (TypeScript).

    - Snowbridge PRs · Polkadot-SDK PRs (Snowbridge pallets run upstream in the Bridge Hub runtime)

- **Polkadot-SDK: Speculative Messaging** | Contributor · 2026 – Present — *Messaging between parachains (ongoing)*

    - **What it is:** The planned successor to HRMP. Senders commit message roots per block, and the relay chain only checks that consumed roots are recent.
    - **My part:** Building the runtime and node sides as one PR stack, from relay-chain checks to end-to-end tests. Primitives are in upstream review (#12676).

- **Parallel Finance** | Blockchain Engineer · 2022 — *DeFi on Polkadot and Ethereum*

    - **Parallel (Rust / Substrate):** A Polkadot parachain with a money market, an AMM, and liquid staking in one runtime. Worked on liquidation in the money market and on on-chain pricing of LP and staking tokens used as collateral. Integrated EVM support and the Axelar bridge.
    - **ParaSpace (Solidity, Ethereum mainnet):** An Aave V3–based market that lends against NFTs and ERC-20s. Worked on NFT liquidation and the NFT floor-price oracle.
    - Parallel PRs · ParaSpace PRs

- **Bifrost Finance** | Blockchain Developer — *Liquid staking on Polkadot (Rust / Substrate)*

    - Worked on the SALP pallet, which gives users a tradable token for DOT locked in parachain crowdloans, and on its contribution, refund, and reward logic.

- **Subscan** | Backend Developer — *Block explorer for Substrate chains (Go)*

    - Worked on the services that index blocks and events into SQL and serve the explorer APIs. Contributed to the open-source subscan-essentials.

Education

- **USTC (University of Science and Technology of China)**
    - M.S. in Information Science & Technology
    - B.S. in Electronic Information Science & Technology

Languages

- English: CET-6. 5+ years working in English-speaking teams and international open-source projects.
- Chinese: native.